

991 + 981 , MY 2012 =>

Diagnosis information - Parking Brake

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000001

Diagnosis information - Control module

Temperature

Diagnostic conditions

The control unit must be active in order to test internal function sequences and diagnosable inputs/outputs.

In the event of a fault, check whether the fault is permanently >present<. To perform the check, erase the fault memory, meet the diagnostic conditions and read out the fault memory again.

- Ignition on

Fault setting condition(s)

- **Device temperature** - Limit exceeded
- **Device temperature** - Below limit

Fault effects

Function not available or available only to a limited extent

Diagnosis information - Control module

Temperature

Diagnostic conditions

The control unit must be active in order to test internal function sequences and diagnosable inputs/outputs.

In the event of a fault, check whether the fault is permanently >present<. To perform the check, erase the fault memory, meet the diagnostic conditions and read out the fault memory again.

- Ignition on

Fault setting condition(s)

- **Device temperature** - Limit exceeded
- **Device temperature** - Below limit

Fault effects

Function not available or available only to a limited extent

000003

Diagnosis information - Spreader device

Diagnostic conditions

- Ignition on

Fault setting condition(s)

Overtemperature **Spreader device** (reversible)

Fault effects

none or Limited function **Parking brake control unit**

000004

Diagnosis information - Parking brake control unit

Spreader device Mechanical fault

Diagnostic conditions

- Ignition on
- Actuation **Parking brake**

Fault setting condition(s)

- Engine blocked
- Short circuit to B+/ground/between lines or open circuit in the lines
- **Spreader device** Faulty

Fault effects

none or Limited function **Parking brake**

Nominal values Line test

Short circuit to B+ :

♦ Setpoint voltage **- 0.5 V** ⇒ **+ 0.5 V**

Short circuit to ground :

♦ Setpoint resistance **> 0.9 MOhm**

Open circuit:

♦ Setpoint resistance **< 2 Ohm**

000005

Diagnosis information - Parking brake control unit

Spreader device Mechanical fault

Diagnostic conditions

- Ignition on
- Actuation **Parking brake**

Fault setting condition(s)

- Engine blocked
- Short circuit to B+/ground/between lines or open circuit in the lines
- **Spreader device** Faulty

Fault effects

none or Limited function **Parking brake**

Nominal values Line test

Short circuit to B+ :

♦ Setpoint voltage **- 0.5 V** ⇒ **+ 0.5 V**

Short circuit to ground :

♦ Setpoint resistance **> 0.9 MOhm**

Open circuit:

♦ Setpoint resistance **< 2 Ohm**

000006

Diagnosis information - Parking brake control unit

Service mode Active

Diagnostic conditions

- Ignition on

Fault setting condition(s)

- **Control module** new
- Loop-in incomplete / Faulty
- No/incorrect calibration

Fault effects

none or Limited function **Parking brake control unit**

Diagnosis information - Clutch sensor

Electrical fault

Diagnostic conditions

The control unit must be active in order to test internal function sequences and diagnosable inputs/outputs.

In the event of a fault, check whether the fault is permanently >present<. To perform the check, erase the fault memory, meet the diagnostic conditions and read out the fault memory again.

- Ignition on
- Actuation Clutch

Fault setting condition(s)

- Short circuit to B+/ground/between lines or open circuit in the lines
- **Clutch sensor** Function restriction

Fault effects

- DAR (DriveAwayRelease) not available
- Roller test bench mode not available

Nominal values Line test

Short circuit to B+ :

♦ Setpoint voltage **- 0.5 V** ⇒ **+ 0.5 V**

Short circuit to ground :

♦ Setpoint resistance **> 0.9 MOhm**

Open circuit:

♦ Setpoint resistance **< 2 Ohm**

Diagnosis information - Clutch sensor

Signal implausible

Diagnostic conditions

The control unit must be active in order to test internal function sequences and diagnosable inputs/outputs.

In the event of a fault, check whether the fault is permanently >present<. To perform the check, erase the fault memory, meet the diagnostic conditions and read out the fault memory again.

- Ignition on
- while driving

Fault setting condition(s)

- The signal of the clutch sensor is outside the value range
- During driving (10 - 80 km/h / 6 - 50 mph), a gear change is recognized based on the monitored transmission ratio, but no corresponding clutch actuation is determined. This state must exist at least 20 times per ignition change for the fault to be entered.
- The vehicle accelerates from standstill to 120 km/h (75 mph) without a clutch actuation being detected. This state must exist 3 times per ignition change for the fault to be entered.

Possible fault causes

- Old Software version **Parking brake control unit** (if a control unit with the part number 991.618.145.02 and a software version < 1600 is installed, please refer to the corresponding symptom-based Workshop Manual)
- Short circuit to B+/ground/between lines or open circuit in the lines
- **Clutch sensor** Function restriction
- **Parking brake control unit** Function restriction

Fault effects

- DAR (DriveAwayRelease) not available

Nominal values Line test

Short circuit to B+ :

♦ Setpoint voltage **- 0.5 V** ⇒ **+ 0.5 V**

Short circuit to ground :

◆ Setpoint resistance > **0.9 MOhm**

Open circuit:

◆ Setpoint resistance < **2 Ohm**

000010

Diagnosis information - Parking brake control unit

Lead Function light Button - Electrical fault

Diagnostic conditions

- Ignition on

Fault setting condition(s)

- Short circuit to B+/ground/between lines or open circuit in the lines
- Faulty

Fault effects

- Function light off

Nominal values Line test

Short circuit to B+ :

♦ Setpoint voltage **- 0.5 V** $\Rightarrow$ **+ 0.5 V**

Short circuit to ground :

♦ Setpoint resistance **> 0.9 MOhm**

Open circuit :

♦ Setpoint resistance **< 2 Ohm**

Diagnosis information - Parking brake control unit

Parking brake control unit - Faulty

Diagnostic conditions

The control unit must be active in order to test internal function sequences and diagnosable inputs/outputs.

In the event of a fault, check whether the fault is permanently >present<. To perform the check, erase the fault memory, meet the diagnostic conditions and read out the fault memory again.

- Ignition on

Possible fault causes

- ♦ Parking brake control unit Faulty

Fault effects

- ♦ none or Limited function **Parking brake**

000012

Diagnosis information - Parking brake control unit

Button Parking brake - Electrical fault

Diagnostic conditions

- Ignition on

Fault setting condition(s)

- Short circuit to B+/ground/between lines or open circuit in the lines
- **Button** Faulty

Fault effects

- Actuation **Parking brake** No function

Nominal values Line test

Short circuit to B+ :

- ♦ Setpoint voltage **- 0.5 V** $\Rightarrow$ **+ 0.5 V**

Short circuit to ground :

- ♦ Setpoint resistance **> 0.9 MOhm**

Open circuit:

- ♦ Setpoint resistance **< 2 Ohm**

000013

Diagnosis information - Parking brake control unit

Fault **Button**

Diagnostic conditions

- Ignition on

Fault setting condition(s)

- Button is stuck or permanently actuated.
- Short circuit to B+/ground/between lines or open circuit in the lines
- Faulty

Fault effects

- Actuation **Parking brake** No function

Nominal values Line test

Short circuit to B+ :

♦ Setpoint voltage **- 0.5 V** $\Rightarrow$ **+ 0.5 V**

Short circuit to ground :

♦ Setpoint resistance **> 0.9 MOhm**

Open circuit:

♦ Setpoint resistance **< 2 Ohm**

Diagnosis information - Parking brake control unit

Parking brake control unit - Faulty

Diagnostic conditions

The control unit must be active in order to test internal function sequences and diagnosable inputs/outputs.

In the event of a fault, check whether the fault is permanently >present<. To perform the check, erase the fault memory, meet the diagnostic conditions and read out the fault memory again.

- Ignition on

Possible fault causes

- ♦ Parking brake control unit Faulty

Fault effects

- ♦ none or Limited function **Parking brake**

Diagnosis information - Parking brake control unit

Parking brake control unit - Electrical fault

Diagnostic conditions

The control unit must be active in order to test internal function sequences and diagnosable inputs/outputs.

In the event of a fault, check whether the fault is permanently >present<. To perform the check, erase the fault memory, meet the diagnostic conditions and read out the fault memory again.

- Ignition on
- **Parking brake** actuated

Possible fault causes

- ♦ **Plug** and / or Contacts damaged
- ♦ Overvoltage / Undervoltage Terminal 30 **Connection**
- ♦ Short circuit to B+/ground/between lines or open circuit in the lines
- ♦ **Parking brake control unit** Faulty
- ♦ **Spreader device** right / left Faulty

Fault effects

- ♦ none or Limited function **Parking brake**

000017

Diagnosis information - Parking brake control unit

Spreader device Faulty

Diagnostic conditions

- Ignition on
- Actuation **Parking brake**

Fault setting condition(s)

- **Spreader device** Faulty

Fault effects

none or Limited function **Parking brake**

Diagnosis information - Parking brake control unit

Calibration

Diagnostic conditions

- Ignition on

Fault setting condition(s)

- Calibration not performed
- Calibration incorrect because clearance of parking brake shoes is not set correctly (too high, too low)
- Calibration incorrect because brakes were not bedded in or not bedded in sufficiently

❗ Following replacement of the **Parking brake control unit** the control unit no longer knows that it has already been bedded in. For this reason, the 'Grind in parking brake' procedure under 'Maintenance/repairs' must also be carried out after replacing the control unit.

Fault effects

none or Limited function **Parking brake control unit**

Diagnosis information - Parking brake control unit

Calibration

Diagnostic conditions

- Ignition on

Fault setting condition(s)

- Calibration not performed
- Calibration incorrect because clearance of parking brake shoes is not set correctly (too high, too low)
- Calibration incorrect because brakes were not bedded in or not bedded in sufficiently

❗ Following replacement of the **Parking brake control unit** the control unit no longer knows that it has already been bedded in. For this reason, the 'Grind in parking brake' procedure under 'Maintenance/repairs' must also be carried out after replacing the control unit.

Fault effects

none or Limited function **Parking brake control unit**

000020

Diagnosis information - Parking brake control unit

Parking brake No function

Diagnostic conditions

- Ignition on
- Actuation **Parking brake**

Fault setting condition(s)

- Overload **Parking brake** (reversible)
- Internal control unit fault (reversible)
- Fault entry **PSM control module** (**Speed sensor**)

Fault effects

Parking brake control unit - Function temporarily unavailable or available only to a limited extent

Diagnosis information - Parking brake control unit

DAR (DriveAwayRelease) not available

Diagnostic conditions

The control unit must be active in order to test internal function sequences and diagnosable inputs/outputs.

In the event of a fault, check whether the fault is permanently >present<. To perform the check, erase the fault memory, meet the diagnostic conditions and read out the fault memory again.

- Ignition on

Fault setting condition(s)

- Doors not closed
- Driver seat belt not fastened

Fault effects

The parking brake does not release automatically when the intention to drive off is detected

Diagnosis information - Parking brake control unit

Calibration

Diagnostic conditions

- Ignition on

Fault setting condition(s)

- Calibration not performed
- Calibration incorrect because clearance of parking brake shoes is not set correctly (too high, too low)
- Calibration incorrect because brakes were not bedded in or not bedded in sufficiently

❗ Following replacement of the **Parking brake control unit** the control unit no longer knows that it has already been bedded in. For this reason, the 'Grind in parking brake' procedure under 'Maintenance/repairs' must also be carried out after replacing the control unit.

Fault effects

none or Limited function **Parking brake control unit**

Diagnosis information - Parking brake control unit

Calibration

Diagnostic conditions

- Ignition on

Fault setting condition(s)

- Calibration not performed
- Calibration incorrect because clearance of parking brake shoes is not set correctly (too high, too low)
- Calibration incorrect because brakes were not bedded in or not bedded in sufficiently

❗ Following replacement of the **Parking brake control unit** the control unit no longer knows that it has already been bedded in. For this reason, the 'Grind in parking brake' procedure under 'Maintenance/repairs' must also be carried out after replacing the control unit.

Fault effects

none or Limited function **Parking brake control unit**

Diagnosis information - Parking brake control unit

Spreader device

Diagnostic conditions

- Ignition on
- Actuation **Parking brake**

Fault setting condition(s)

- Maximum difference reached

ⓘ This fault entry can be an indication of worn-out parking-brake shoes.

Fault effects

The parking brake can only be released but no longer applied. The actuators automatically move to assembly position for renewed adjustment/calibration.

Diagnosis information - Parking brake control unit

Parking brake No function

Diagnostic conditions

- Ignition on
- Actuation **Parking brake**

Fault setting condition(s)

- Overload **Parking brake** (reversible)
- Internal control unit fault (reversible)
- Fault entry **PSM control module** (**Speed sensor**)

Fault effects

Parking brake control unit - Function temporarily unavailable or available only to a limited extent

Diagnostic information - parking brake control unit

Emergency braking via parking brake not available

① During emergency braking using the parking brake, the parking brake button is pressed and held when travelling at a speed of more than 5 km/h (3 mph). The parking brake control unit then sends a deceleration request to the PSM control unit, which then initiates braking using the PSM hydraulic system. As soon as the vehicle speed falls below a certain limit value (approx. 2 km/h or 1.5 mph), the parking brake is closed and the vehicle is brought to a standstill.

Diagnostic conditions

- Emergency braking performed using the parking brake (see above)

Fault setting condition(s)

- Emergency braking function could not be performed

Possible fault causes

- ♦ Other fault memory entries in the parking brake control unit or PSM control unit that prevent braking using PSM/the parking brake

Fault effects

- ♦ Emergency braking function is not available
- ♦ Warning in instrument cluster

Diagnostic information - parking brake control unit

Spreader device (actuator) in intermediate position

Diagnostic conditions

- Ignition on
- **Parking brake** actuation
- No actively present fault triggering a parking brake warning light

Fault setting condition(s)

- Activation of the actuator is interrupted without the actuator end position being reached

Possible fault cause(s)

- ♦ Undervoltage during actuator activation
- ♦ Diagnostic request to the parking brake control unit during actuator activation
- ♦ Internal actuator fault (temporary/reversible)

❗ The fault is reversible in many cases, and it can be corrected by opening/closing the parking brake several times by means of the button element or diagnostics.

Fault effects

No direct effects/fault is reversible

Diagnosis information - Parking brake control unit

Calibration

Diagnostic conditions

- Ignition on

Fault setting condition(s)

- Calibration not performed
- Calibration incorrect because clearance of parking brake shoes is not set correctly (too high, too low)
- Calibration incorrect because brakes were not bedded in or not bedded in sufficiently

❗ Following replacement of the **Parking brake control unit** the control unit no longer knows that it has already been bedded in. For this reason, the 'Grind in parking brake' procedure under 'Maintenance/repairs' must also be carried out after replacing the control unit.

Fault effects

none or Limited function **Parking brake control unit**

000030

Diagnosis information - Parking brake control unit

Spreader device Electrical fault

Diagnostic conditions

- Ignition on
- Actuation **Parking brake**

Fault setting condition(s)

- Short circuit to B+/ground/between lines or open circuit in the lines
- Undervoltage
- **Spreader device** Faulty

Fault effects

none or Limited function **Parking brake**

Nominal values Line test

Short circuit to B+ :

♦ Setpoint voltage **- 0.5 V** ⇒ **+ 0.5 V**

Short circuit to ground :

♦ Setpoint resistance > **0.9 MOhm**

Open circuit:

♦ Setpoint resistance < **2 Ohm**

000031

Diagnosis information - Parking brake control unit

Spreader device Electrical fault

Diagnostic conditions

- Ignition on
- Actuation **Parking brake**

Fault setting condition(s)

- Short circuit to B+/ground/between lines or open circuit in the lines
- Undervoltage
- **Spreader device** Faulty

Fault effects

none or Limited function **Parking brake**

Nominal values Line test

Short circuit to B+ :

♦ Setpoint voltage **- 0.5 V** ⇒ **+ 0.5 V**

Short circuit to ground :

♦ Setpoint resistance > **0.9 MOhm**

Open circuit:

♦ Setpoint resistance < **2 Ohm**

Diagnosis information - Parking brake control unit

Calibration

Diagnostic conditions

- Ignition on

Fault setting condition(s)

- Calibration not performed
- Calibration incorrect because clearance of parking brake shoes is not set correctly (too high, too low)
- Calibration incorrect because brakes were not bedded in or not bedded in sufficiently

❗ Following replacement of the **Parking brake control unit** the control unit no longer knows that it has already been bedded in. For this reason, the 'Grind in parking brake' procedure under 'Maintenance/repairs' must also be carried out after replacing the control unit.

Fault effects

none or Limited function **Parking brake control unit**

Diagnosis information - Parking brake control unit

Calibration

Diagnostic conditions

- Ignition on

Fault setting condition(s)

- Calibration not performed
- Calibration incorrect because clearance of parking brake shoes is not set correctly (too high, too low)
- Calibration incorrect because brakes were not bedded in or not bedded in sufficiently

❗ Following replacement of the **Parking brake control unit** the control unit no longer knows that it has already been bedded in. For this reason, the 'Grind in parking brake' procedure under 'Maintenance/repairs' must also be carried out after replacing the control unit.

Fault effects

none or Limited function **Parking brake control unit**

000034

Diagnosis information - Parking brake control unit

Spreader device Mechanical fault

Diagnostic conditions

- Ignition on
- Actuation **Parking brake**

Fault setting condition(s)

- Engine blocked
- Short circuit to B+/ground/between lines or open circuit in the lines
- **Spreader device** Faulty

Fault effects

none or Limited function **Parking brake**

Nominal values Line test

Short circuit to B+ :

♦ Setpoint voltage **- 0.5 V** ⇒ **+ 0.5 V**

Short circuit to ground :

♦ Setpoint resistance **> 0.9 MOhm**

Open circuit:

♦ Setpoint resistance **< 2 Ohm**

000035

Diagnosis information - Parking brake control unit

Spreader device Mechanical fault

Diagnostic conditions

- Ignition on
- Actuation **Parking brake**

Fault setting condition(s)

- Engine blocked
- Short circuit to B+/ground/between lines or open circuit in the lines
- **Spreader device** Faulty

Fault effects

none or Limited function **Parking brake**

Nominal values Line test

Short circuit to B+ :

♦ Setpoint voltage **- 0.5 V** ⇒ **+ 0.5 V**

Short circuit to ground :

♦ Setpoint resistance **> 0.9 MOhm**

Open circuit:

♦ Setpoint resistance **< 2 Ohm**

000036

Diagnosis information - Parking brake control unit

Spreader device Faulty

Diagnostic conditions

- Ignition on
- Actuation **Parking brake**

Fault setting condition(s)

- **Spreader device** Faulty

Fault effects

none or Limited function **Parking brake**

Diagnosis information - Parking brake control unit

Spreader device

Diagnostic conditions

- Ignition on
- Actuation **Parking brake**

Fault setting condition(s)

- Maximum difference reached

ⓘ This fault entry can be an indication of worn-out parking-brake shoes.

Fault effects

The parking brake can only be released but no longer applied. The actuators automatically move to assembly position for renewed adjustment/calibration.

000038

Diagnosis information - Parking brake control unit

Spreader device Electrical fault

Diagnostic conditions

- Ignition on
- Actuation **Parking brake**

Fault setting condition(s)

- Short circuit to B+/ground/between lines or open circuit in the lines
- Undervoltage
- **Spreader device** Faulty

Fault effects

none or Limited function **Parking brake**

Nominal values Line test

Short circuit to B+ :

♦ Setpoint voltage **- 0.5 V** ⇒ **+ 0.5 V**

Short circuit to ground :

♦ Setpoint resistance **> 0.9 MOhm**

Open circuit:

♦ Setpoint resistance **< 2 Ohm**

000040

Diagnosis information - Parking brake control unit

Lead Lamp Instrument cluster - Electrical fault

Diagnostic conditions

- Ignition on

Fault setting condition(s)

- Short circuit to B+/ground/between lines or open circuit in the lines

Fault effects

- none or **Function light** off

Nominal values Line test

Short circuit to B+ :

♦ Setpoint voltage **-0.5 V** $\Rightarrow$ **+0.5 V**

Short circuit to ground :

♦ Setpoint resistance **> 0.9 MOhm**

Open circuit:

♦ Setpoint resistance **< 2 Ohm**

000041

Diagnosis information - Parking brake control unit

Terminal 15 - Signal implausible

Diagnostic conditions

- Ignition on

Possible fault causes

- ♦ The signal of the terminal 15 connected to the control unit does not correspond to the terminal 15 signal via CAN.

Fault effects

- ♦ none